

Sensor XT MVP SSI MLT 20 C BT

Sensor for traffic, presence and parking detection

Product description

- Award-winning multifunctional sensor with camera sensor and DALI interface for integration into street lighting
- High level of data protection as images are only processed within the sensor
- Detection of people and vehicles
- Reliable detection on different weather conditions
- Minimal false detection thanks to artificial intelligence algorithms
- Up to 31 detection zones can be configured with the associated app via Bluetooth
- Traffic density measurement (vehicles per time)
- Presence detection of people in freely definable areas
- Notifications of car park occupancy
- Light measurement
- All measured data is available via DALI instances
- Application controller for controlling the light
- Ready for Zhaga Book 18 Ed. 3 receptacle for easy and flexible installation to luminaire
- Lifetime up to 100,000 h at $t_c = 60^\circ\text{C}$
- Guarantee conditions are defined separately by project
- We will provide security updates at least till 30th November 2026



Housing properties

- Casing: Dark grey (RAL 7031)
- Type of protection IP66
- Impact protection degree IK08

Benefits

- **Multifunction:** One sensor can cover different sensing applications
- **Flexible:** Detection zone and the artifact to be detected can be defined by the user
- **Scalable:** The sensor is able to work from single luminaires up to CMS systems

Typical applications

- Light levels adjustment according to real time traffic patterns
- Traffic data collection
- Presence detection on crosswalks and sidewalks
- On and off street parking lots management



Standards, page 4

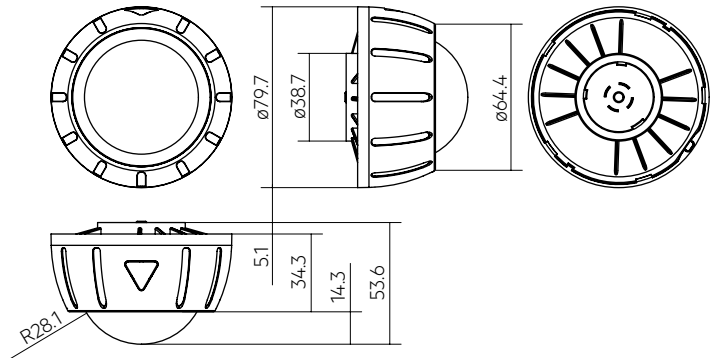
Wiring diagrams and installation examples, page 5

Sensor XT MVP SSI MLT 20 C BT

Sensor for traffic, presence and parking detection

Technical data

Sensor type	Edge based vision system based on artificial intelligence
Supply voltage ^①	24 V
Current draw	41 mA
Max. power consumption	1 W
Starting time ^②	110 – 1,200 ms
Supported operating modes	128
Mounting height	4.5 – 12 m
Mounting hole	acc. to Zhaga Book 18 Ed. 3
Type of installation ^③	Zhaga Book 18 Ed. 3 socket
Field of view ^⑥	ø24.1 m
Detection angle for light measurement in X direction	60°
Detection angle for light measurement in Y direction	50°
Detection range for light measurement ^④	1 – 32,512 lx
Radio transceiver operating frequencies	2.4 – 2.483 GHz
Max. output power radio transceiver (E.R.I.P.) ^⑤	< +8 dBm
Radio protocol	Bluetooth 5.2
Max. speed of vehicles ^⑧	45 – 80 km / h
Ambient temperature ta ^⑦	-25 ... +50 °C
tc	60 °C
Storage temperature	-25 ... +60 °C
Humidity	0 – 90 %
Dimensions Ø x H	ø79.7 x 53.6 mm
Housing material body	Polycarbonate
Housing material lens	Polycarbonate
Housing colour body	Dark grey (RAL 7031)
Housing colour lens	Crystal clear
Type of protection	IP66
Impact protection degree ^⑨	IK08
Lifetime	up to 100,000 h
Guarantee (Conditions at www.tridonic.com)	2 years



Ordering data

Type	Article number	Suitable for	Packaging carton	Weight per pc.
Sensor XT MVP SSI MLT 20 C BT	28005152	Street & Road lighting	20 pc(s).	0.127 kg

① Only AUX power supply as specified by DiiA part 351. Voltage range 21.6 – 26.4 V.

② Starting time for DALI communication. It takes approximately 15 s for the sensor to deliver effective data.

③ Device must be installed on the bottom side of the luminaire.

④ Value measured at the sensor head. This corresponds to 20 – 90 % of the lux value measured on the surface located below the sensor. If luminaire goes to stand-by and there is no other light source detection is not working.

⑤ E.I.R.P.: Equivalent Isotropically Radiated Power.

⑥ This value depends on the installation height. Details see data sheet chapter 5.5.

⑦ Correct sensor operation can not be guaranteed if operated outside this ta window.

⑧ It is essential to mount the plug connection (AZU Z18 Set) and sensor correctly to achieve the full IK rating. These impact protection degree is applicable for all parts of the housing including optics.

ACU Z18 Set (Zhaga Book 18 Ed. 3 receptacle)

Product description

- Set consisting of receptacle, lock washer and mounting nut (ACU Z18 REP 4PIN)
- Designed for street lighting control
- Compliant with Zhaga Book 18 Ed. 3
- Optional IP66 with sealing cap (ACU Z18 CAP IP66)
- UL UV-f1 rated for outdoor use
- IK09 high impact resistant
- For more details see data sheet ACU Z18 Set



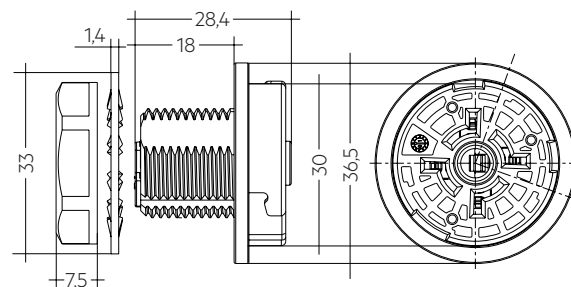
Housing properties

- Casing receptacle: PBT
- Type of protection IP20

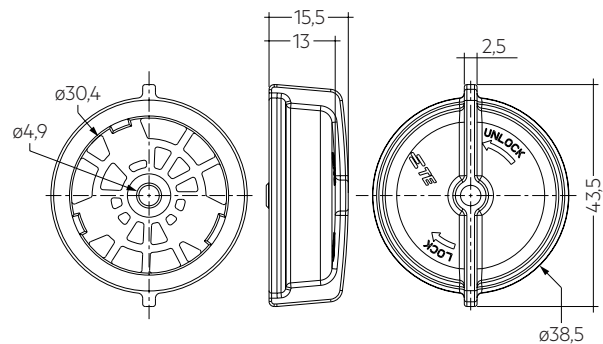
Interfaces

- Supplies Zhaga Book 18 Ed. 3 contact system

ACU Z18 REP 4PIN + ACU Z18 CAP IP66



ACU Z18 REP 4PIN



ACU Z18 CAP IP66

Ordering data

Type	Article number	Packaging bag	Weight per pc.
ACU Z18 REP 4PIN	28003209	100 pc(s).	0.008 kg
ACU Z18 CAP IP66	28003208	100 pc(s).	0.020 kg

1. Standards

EN 61347-1
EN 61347-2-11
EN 55015
EN 61547
EMC directive 2014/30/EC
EN 300 328
EN 301 489-1
EN 301 489-17
EN 62479
EN 62493

1.2 Glow wire test

according to EN 61347-1 passed with 850 °C.

2. Common

The Tridonic Sensor XT is one of the first camera sensors developed especially to fit perfectly in Urban Outdoor Applications.

With its implemented camera the Sensor XT covers a wide detection range.

With installation heights from 4.5 m up to 12 m the detection area can be up to $\approx \varnothing 24.1$ m.

Functionalities supported are traffic monitoring, high end presence detection and parking lot management.

The innovative platform design of the sensor enables easy plug and play by supporting the Zhaga Book 18 Ed. 3.

Next to its robust design and long lifetime the IP66 rating complements the excellent protection against harsh outdoor environments.

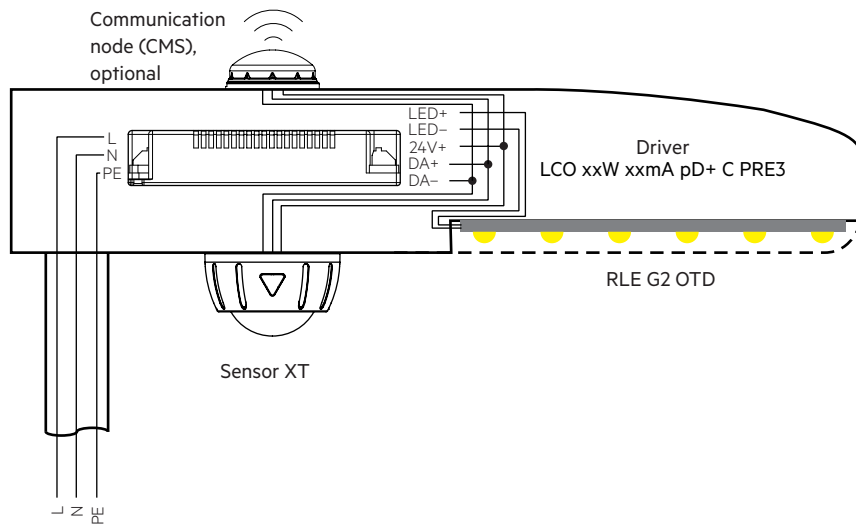
3. Installation

- The Sensor XT must not be connected to mains. It is supplied directly via the 24V AUX power supply.
- DALI is not SELV. The installation instructions for mains voltage therefore apply.
- If installed at another height than specified the sensor is very likely to operate not as intended (e.g. false detection, no detection, etc.).
- The triangular sign indicates the direction to the street (Zhaga y axis) when the sensor is locked in place.
- Make sure the sensor is mounted correctly and locked in place.
- Motions of the lighting pole may trigger false detection or prohibit correct detection of configured detection zones.
- Proper detection might be impacted by obstacles blocking the direct line of sight between camera and detection zone. e.g. non-stationary signs, automated yard-gates, construction vehicles, etc.
- Avoid direct illumination of the light source on the sensor including housing.
- For proper operation of the sensor, it needs be ensured, that the illuminance level is at least 10 lux (measured at the sensor head) within the whole detection area at all times. If luminaire goes to stand-by and there is no other light source detection is not working.
- Fewer but larger detection zones result in better detection accuracy.

3.1 Wiring

OTD system overview:

The sensor is optimized to be used in combination with Tridonic LED driver LCO PRE3 with internal power supply (24 V AUX power supply is mandatory). Following illustration visualizes the wiring of the components inside the luminaire.



The max. cable length is 2 m.

The sensor supports the Zhaga connectivity standard for Plug and Play luminaire extensions. This allows a toolless and flexible integration in luminaires. The connection interface is designed according to Zhaga Book 18 Ed. 3.

Pin assignment Sensor XT

Pins	Assignment
Pin 1	+ 24 V
Pin 2*	DA- / ground for 24 V and general Digital I/O
Pin 3*	DA+
Pin 4	General I/O, not connected

* Correct polarity is mandatory for regular operation of the sensor.

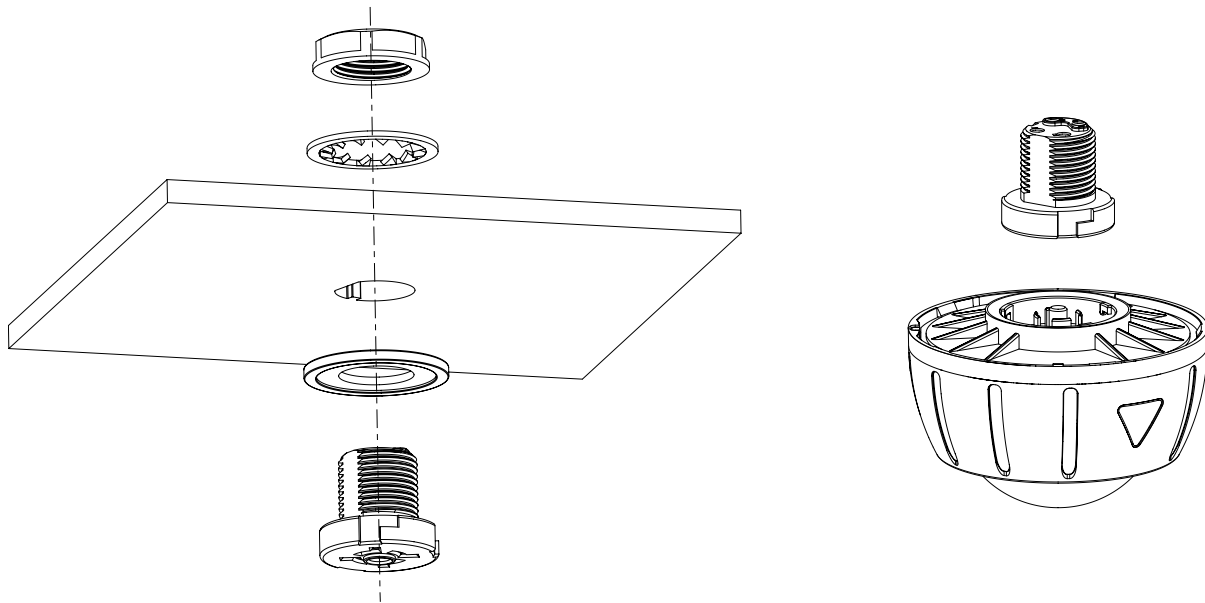
The receptacle has to be designed according to Zhaga Book 18 Ed. 3 and inside the luminaire the polarity has to be wired correctly.



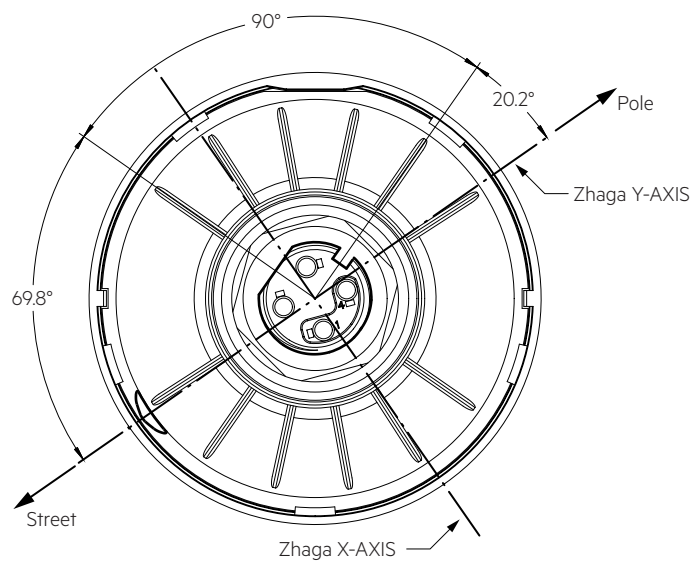
The Zhaga connector pins mounted in the luminaire, have to follow the same pin assignment as described in Zhaga Book 18 Ed. 3!

3.2 Mounting instructions

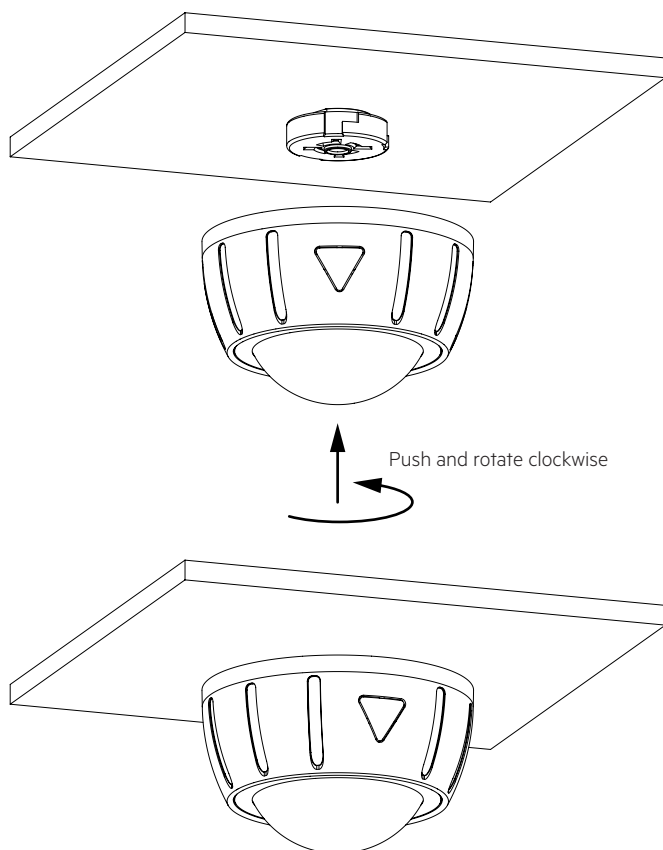
Assembly visualization, see data sheet AZU Z18 Set too:



Zhaga Book 18 Ed. 3 axis description:



Installation on luminaire:



The triangular sign indicates the direction to the street (Zhaga y axis) when the sensor is locked in place.
The sensor must lock.

4. Software / Programming / Interfaces

4.1 Software / programming

With appropriate software and interface different functions can be activated and various parameters can be configured in the sensor. The sensor supports the following software and interfaces:

Software / hardware for configuration:

- Tridonic Sensor XT mobile app

Interface for data transfer:

- Control input DALI

Interface for commissioning:

- Bluetooth

5. Sensor function

The Sensor XT acts as a DALI-input-device with multiple instances. Please also refer to the manual. It contains 32 DALI instances. These are structured as follows:

DALI instances			
Instance number	Instance type	Explanation	Reference applicable standard
0	4	Light Sensor	IEC 62386-304 Ed.1
1	3	Presence Detector Zone 1	IEC 62386-303 Ed.1
2	3	Presence Detector Zone 2	IEC 62386-303 Ed.1
...	...	...	...
20	3	Presence Detector Zone 20	IEC 62386-303 Ed.1
21	6	General Purpose Sensor Zone 1	IEC 62386-306 Ed.1
22	6	General Purpose Sensor Zone 2	IEC 62386-306 Ed.1
...	...	...	...
31	6	General Purpose Sensor Zone 11	IEC 62386-306 Ed.1

There are 2 different operation types:

- Application Controller (Standalone Operation):
Integrated MultiMaster Application Controller is used.
Only local control of the directly connected DALI drivers is possible.
All connected drivers react identical (DALI Broadcast).
Functionality in this mode is switching on/off light based on presence (either people or cars) or dimming light to preset levels depending on traffic density (number of cars / time).
- Monitoring Detection (Sensor Operation in combination with a CMS):
Processing sensor information by separate MultiMaster ApplicationController.

There are three different use cases for the Sensor XT.

5.1 Traffic Monitoring

Applications:

- Traffic observation
- Traffic diagnostics
- Adaption of light level based on traffic information

Reported values:

- Vehicles per time unit (min or h)

5.2 High End Presence Detection

Applications:

- Detection of persons
- Presence detection for scenarios which require specific detection zones which can't be implemented with standard sensors
- Presence detection for scenarios which require multiple overlapping / non-overlapping detection zones
- Adaption of light level based on presence
- Attention zones (e.g. no step zones in train stations)

Reported values:

- Object presence in a single detection zone (yes / no)

5.3 Parking Lot Management

Applications:

- Parking lot occupancy (On / Off Street parking)

Reported values:

- Status per parking lot (free / occupied)

5.4 Conditions for accurate presence detection

Sensor XT is an optical image sensor. Therefore environmental conditions impacting the visibility of objects may influence the reliability of the detection. The following exemplary influencing factors may occur in combination:

- reduced contrast levels
- reduced optical visibility through e.g. dust, smoke, clouds
- extreme glare e.g. by reflexions of sunlight in windows
- (temporary) object impacting direct line of sight to detection area. e.g. a parked truck

Please note that influencing factors may be permanent or temporary.

Due to optical setup the resolution is higher in middle of picture, compared to edge area of picture.

The bigger the object as better the detection in the edge area.

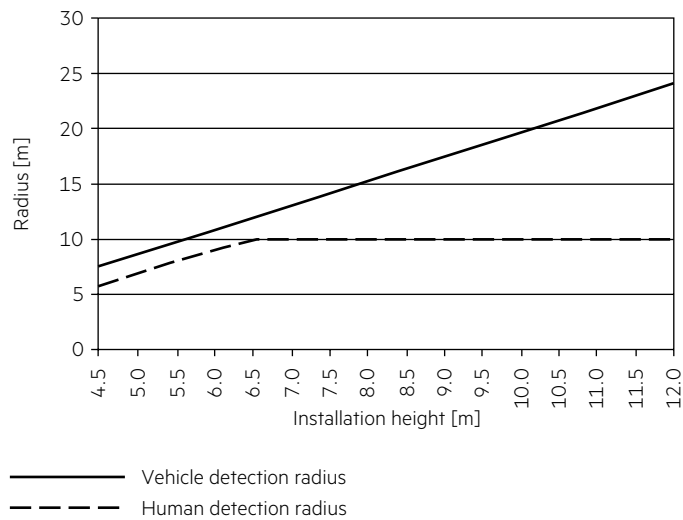
For example cars can be detected better than people in the edge area.



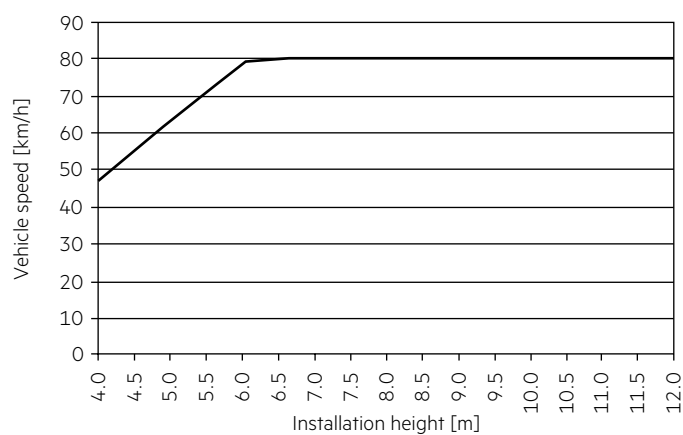
The sensorMODE of Tridonic LED driver LCO PRE3 should not be used with Sensor XT.
Either use internal application controller or processing sensor info via external CMS.

5.5 Motion detection area and object speed

The size of the detection area changes depending on the mounting height of the Sensor XT and the object to be detected.



The maximum speed of the objects to be detected is limited to 80 km/h. It is also dependent on the installation height of the Sensor XT.



Height	Detection radius vehicle	Detection radius human	Vehicle speed
4,5 m	7,6 m	5,9 m	55 km/h
6,0 m	10,9 m	9,2 m	79 km/h
7,5 m	10,2 m	10,0 m	80 km/h
9,0 m	17,5 m	10,0 m	80 km/h
10,5 m	20,8 m	10,0 m	80 km/h
12,0 m	24,1 m	10,0 m	80 km/h

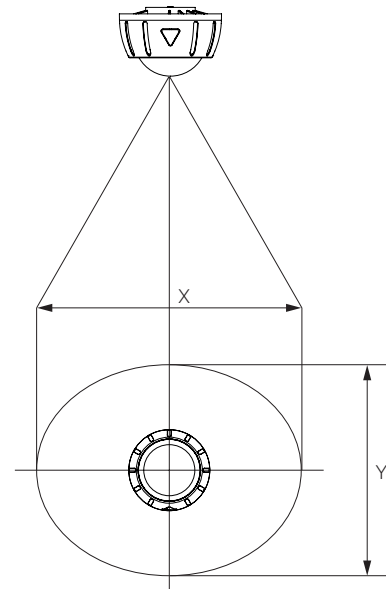
5.6 Light measurement

The light measurement has a cone-shaped detection area with a angle of 60° in the x-direction and 50° in the y-direction.

The measurement range is between 1 and 32,512 lux. Measured at the sensor head.

The measured value is limited by the DALI-2 protocol.

To measure the given range, the sensor mode 128 needs to be used.



Height	dx	dy
4,5 m	5,2 m	4,2 m
6,0 m	6,9 m	5,6 m
7,5 m	8,7 m	7,0 m
9,0 m	10,4 m	8,4 m
10,5 m	12,1 m	9,8 m
12,0 m	13,9 m	11,2 m

5.6.1 Light value format

The operating mode is 128.

For the light measurement floating point values are used.

5.7 Status LED

There is a single red LED built-in to indicate the status information to the user.

- LED flashes 5x when power is applied to the device to indicate proper supply
- When the DALI-2 command "Identify device" is sent to the device, the LED flashes for 10 seconds

6. Miscellaneous

6.1 Disposal of equipment



Return old devices in accordance with the WEEE directive to suitable recycling facilities.

6.2 Additional information

Additional technical information at www.tridonic.com → Technical Data

Guarantee conditions at www.tridonic.com → Services

Lifetime declarations are informative and represent no warranty claim.
No warranty if device was opened.